ModelIt!

Student Activity Pack

Coral Reef Collapse: Death of an Underwater City

Modeling Cascading Ecosystem Failure Through Coupled Feedback Loops

Grade 09L2 | NGSS HS-LS2-6, HS-ESS3-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What causes coral to turn white and why is that a problem?

2 How can a tiny temperature change destroy an entire ecosystem?

3 Draw a diagram showing relationships between coral, algae, fish, and water temperature.

4 What does tipping point mean in an ecosystem?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Ocean Temperature | Coral Bleaching Rate | Algae Overgrowth | Fish Population | Water Acidity | Symbiotic Zooxanthellae | Reef Structural Integrity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mild Warming

Temperature +0.5 C for one season

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sustained Warming

Temperature +1.5 C maintained

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Triple Threat

Temperature +1 C, acidity +0.1 pH, high fishing

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Coral Recruitment Rate — Rate at which new coral larvae settle and grow, determining reef regeneration ability

Sedimentation Rate — Amount of sediment from coastal development that smothers coral and blocks light

Marine Protected Area Coverage — Percentage of reef protected from fishing and development

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a 1 degree C temperature change destroy an ecosystem that took 10,000 years to build?
Use evidence from your simulation to support your answers.

1 How does your model demonstrate that reef collapse is a cascading system failure?

2 Why is the phase shift so difficult to reverse even if temperatures stabilize?

3 How do multiple moderate stressors interact differently than a single severe stressor?

4 What does your model suggest about conservation vs. restoration?

5 How might reef loss affect coastal human communities?

8 STEM Challenge: Design a Reef Resilience Monitoring System

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YOUR ENGINEERING MISSION

Design a monitoring system that detects early warnings of reef collapse and triggers interventions before the phase shift becomes irreversible.

THE SCENARIO

The Great Barrier Reef has experienced three mass bleaching events in five years. Design an early-warning system that predicts which reef sections are approaching the tipping point.

Engineering Design Thinking

1. WHAT SENSORS AND MONITORING TECHNOLOGIES WOULD YOU DEPLOY?
2. HOW WOULD YOU BUILD A PREDICTIVE MODEL FOR AT-RISK REEF SECTIONS?
3. HOW WOULD YOU TRIAGE INTERVENTIONS ACROSS HUNDREDS OF SECTIONS?
4. HOW WOULD YOU VALIDATE YOUR INTERVENTIONS ARE PREVENTING PHASE SHIFTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Marine Ecologists study ocean ecosystem dynamics and conservation, working for NOAA, marine parks, and environmental NGOs, earning $55,000-$110,000/year.